Sequence & Progression

Mind map

Arithmetic progression

- $T_n = a + (n-1)d,\ S_n = \frac{n}{2}(a+l)$
- $T_n = S_n - S_{n-1};\ S_n = An^2 + Bn$
- Assume $a-d, a, a+d$

AP highlights

- Equidistant terms sum to $a+l$
- Common terms: $d = \text{lcm}$
- Multiples: inclusion–exclusion

Arithmetic mean

- $A = \frac{a+b}{2}$
- $d = \frac{b-a}{n+1}$
- $\sum A_k = nA$

Geometric progression

- $T_n = ar^{n-1}$
- $S_n = \frac{a(r^n-1)}{r-1}$
- $S_\infty = \frac{a}{1-r},\ |r| < 1$

GP highlights

- $\frac{a}{r}, a, ar\ ;\ b^2 = ac$
- AP $\leftrightarrow$ GP by c^x / log
- $a_{n+1} = pa_n + q$: add k

Geometric mean

- $G = \sqrt{ab}$
- $r = (b/a)^{1/(n+1)}$
- $\prod G_k = (ab)^{n/2}$

AM-GM

- AM $\geq$ GM, equality: all equal
- Min of sum: product constant
- Max of product: sum constant

HP and HM

- Reciprocals in AP
- $H = \frac{2ab}{a+b}$
- $\frac{1}{H_k} = \frac{1}{a} + kd$

AM, GM, HM relation

- RMS $\geq$ AM $\geq$ GM $\geq$ HM
- $G^2 = AH$
- $x^2 - 2Ax + G^2 = 0$

AGP

- $T_n = [a + (n-1)d]r^{n-1}$
- $S - rS$ and repeat
- $S_\infty = \frac{a}{1-r} + \frac{dr}{(1-r)^2}$

Type 1: power sums

- $\sum n,\ \sum n^2,\ \sum n^3$
- Alternating signs: pair terms
- Pattern $9, 99, 999$: $10^n - 1$

Type 2

- Differences in AP / GP
- $T_n = u_1 + \sum(u_r - u_{r-1})$
- Then use type 1

Type 3: method of difference

- Denominator factors in AP
- $T_r = f(r) - f(r-1)$
- Cancel; keep end pieces

Type 4 and continued products

- Numerator: product of r factors
- Add next factor, drop previous
- $T_r = c[P_{r-1} - P_r]$

Recurrence and cubic roots

- $p = \sum a_n x^n$, solve for p
- AP: middle root $-\frac{b}{3a}$
- GP: $(\text{middle})^3 = -\frac{d}{a}$

Groups and applications

- Group sizes $\rightarrow$ first term $\rightarrow$ sum
- Ball: $H + \frac{2He}{1-e}$
- Squares sum $= 0 \Rightarrow$ each $= 0$